



ASSIGNMENT - 07

Deep Learning Basics

DATA SCIENCE

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DEEP LEARNING BASICS

1. What is deep learning and how does it differ from traditional machine learning?
2. What is a neural network (artificial neural network)?
3. What is a perceptron, and how is it related to logistic regression?
4. What are weights and biases in a neural network?
5. Why do we need activation functions in a neural network? What would happen if you had no activation function in hidden layers?
6. What is forward propagation in a neural network?
7. What is a loss (cost) function in the context of neural networks?
8. What is backpropagation and how does it work to adjust weights in a neural network?
9. Why should the initial weights of a neural network be set randomly (rather than all to the same value)?
10. What are vanishing gradients and exploding gradients in deep learning?
11. What is overfitting in the context of training neural networks?
12. Name two techniques to help prevent overfitting in deep learning models.
13. What is dropout and how does it help prevent overfitting?
14. What is batch normalization and why is it useful in training deep networks?
15. What is an epoch in the training process of a neural network?
16. What is the difference between batch gradient descent and stochastic gradient descent?
17. What is a mini-batch in the context of training neural networks?
18. What is the learning rate in training a neural network, and why is it important?
19. Name two optimization algorithms used in deep learning besides basic SGD (stochastic gradient descent).
20. What are some common activation functions used in neural networks? (Give at least two examples.)
21. Why is the ReLU activation function often used in hidden layers?
22. Why are GPUs commonly used for deep learning tasks?
23. Why is it important to monitor both training loss and validation loss during training? What might it indicate if validation loss starts increasing while training loss continues to decrease?
24. What is the difference between a shallow neural network and a deep neural network?
25. What is early stopping during model training?
26. How can you tell if a model is underfitting, and what is one way to increase its capacity?
27. If a neural network's training loss is not decreasing at all, what are some things you would check or try to fix?
28. What is the purpose of having a validation set during training?
29. What activation function would you use in the output layer for a binary classification? What about for a multi-class classification?
30. What is the difference between a loss function and an optimizer in the context of training a neural network?